

RECAP HOMEWORK

Pseudocode Recap

Combining read, write, assignment, if, for, while, and do...while in simple problems

These problems mix everything covered so far: reading and writing data, assignment, the if condition, and all three repetitive structures (for, while, and do...while). Each exercise tells you which structures it expects you to combine — but if you find a cleaner way to solve it using a different structure, that's a good sign you understand the material even better.

Exercise 1 — Sum of the First n Numbers

STRUCTURES: *read · assignment · for · write*

Read a positive integer n . Compute and display the sum $1 + 2 + 3 + \dots + n$.

Exercise 2 — Multiplication Table

STRUCTURES: *read · for · write*

Read an integer n between 1 and 10. Display its multiplication table, from $n \times 1$ up to $n \times 10$.

Exercise 3 — Maximum of n Numbers

STRUCTURES: *read · for · if · assignment · write*

Read a positive integer n , then read n numbers one by one. Display the largest value among them.

Hint: keep a variable holding the largest value seen so far, and update it with an if each time you read a new number.

Exercise 4 — Count Even and Odd Numbers

STRUCTURES: *read · for · if · assignment · write*

Read a positive integer n , then read n integers. Display how many of them are even and how many are odd.

Exercise 5 — Prime Number Check

STRUCTURES: *read · while · if · assignment · write*

Read an integer n greater than 1. Determine and display whether n is a prime number.

Hint: test divisors starting from 2 upward; you can stop early as soon as you find one that divides n evenly, or once you pass $\sqrt{n}$.

Exercise 6 — Running Total Until Zero

STRUCTURES: *do...while · if · assignment · write*

Repeatedly read integers from the user, adding each one to a running total, until the user enters 0. Display the final total.

Hint: a sentinel-controlled loop like this is a natural fit for do...while, since you must read a value at least once before you can check whether it's the sentinel.

Exercise 7 — Validated Grade & Letter

STRUCTURES: *do...while · if · write*

Using do...while, read a grade and keep asking again until the user enters a value between 0 and 100. Then, using if...then...else, display the corresponding letter grade: A for 90–100, B for 80–89, C for 70–79, D for 60–69, and F below 60.

Exercise 8 — FizzBuzz

STRUCTURES: *read · for · if · write*

Read a positive integer n . Using a for structure, display the numbers from 1 to n . For multiples of 3, display “Fizz” instead of the number; for multiples of 5, display “Buzz”; for multiples of both 3 and 5, display “FizzBuzz”.

Exercise 9 — Smallest and Largest in a Sequence

STRUCTURES: *do...while · if · assignment · write*

Read a sequence of positive integers, one at a time, stopping as soon as the user enters -1. Display the smallest and the largest value read (not counting -1).

Exercise 10 — Factorial with Input Validation

STRUCTURES: *do...while · for · if · assignment · write*

Using do...while, read an integer n and keep asking again until n is greater than or equal to 0. Then, using a for structure, compute and display $n!$ (n factorial).

Exercise 11 — Greatest Common Divisor

STRUCTURES: *read · while · assignment · write*

Read two positive integers a and b . Using a while structure and the Euclidean algorithm — repeatedly replace the larger number with the remainder of dividing it by the smaller, until one of them becomes 0 — compute and display their greatest common divisor.